Jaiaid Mobin

Graduate Research Assistant

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https://scholar.google.com/citations?user=ifmzYdsAAAAJ

Summary

5th-year Ph.D. student, working under Dr. M. Mustafa Rafique on application state management to improve memory resource utilization, with two first-authored peer-reviewed publications (one poster, one full paper) and **collaborations on I/O optimization, memory tier management, and autonomous vehicle perception**. Check my GitHub to see some of my work. Looking for an opportunity as a software engineer, performance engineer, or research engineer to start in Spring 2026-27.

Education

Rochester Institute of Technology — *Rochester, NY, USA*
Ph.D. in Computer Science — *August, 2022 - Present*
- **Research Topic** — GPU context management, DDL data management, Checkpoint-Restart Engine
- **Tentative Thesis Title** — Towards Effective Memory Footprint Reduction of DNN Workloads

Bangladesh University of Engineering and Technology — *Dhaka, Bangladesh*
M.Sc. in Computer Science — *August, 2022*
- **Thesis Title** — Transit Network Design For Evacuation Modeling Using Heuristic and Multi-Objective Optimization Based Approaches

Bangladesh University of Engineering and Technology — *Dhaka, Bangladesh*
B.Sc. in Computer Science — *September, 2017*

Selected Publications

- **Mobin, Jaiaid**, Avinash Maurya, M. Mustafa Rafique, and Bogdan Nicolae, **"Unipipe: Unified Inference Serving and Fine-Tuning for Deadline-Constrained DNN Surrogate Workflows"**. IEEE eScience 2026.
- Ali Khalid, **Jaiaid Mobin**, Sumanth Rao Appala, Avinash Maurya, Stephany Berrio Perez, M. Mustafa Rafique, and Fawad Ahmad.**"Been There, Scanned That: Nostalgia-driven LiDAR Compression for Self-Driving Cars"**, ACM/IEEE SenSys 2026.
- Li, Yuze, Shunyu Yao, **Jaiaid Mobin**, Tianyu Zhan, M. Mustafa Rafique, Dimitrios Nikolopoulos, Kirshanthan Sundararajah, and Ali R. Butt. **"Memory Tiering in Python Virtual Machine"**. In Proceedings of the 17th ACM SIGPLAN International Workshop on Virtual Machines and Intermediate Languages, pp. 33-44. 2025.
- **Mobin, Jaiaid**, Avinash Maurya, and M. Mustafa Rafique.**"COLTI: Towards Concurrent and Co-located DNN Training and Inference"**. In Proceedings of the 32nd International Symposium on High-Performance Parallel and Distributed Computing, pp. 309-310. 2023.
- Maurya, Avinash,**Jaiaid Mobin**, and M. Mustafa Rafique. **"Towards data gravity and compliance aware distributed deep learning on hybrid clouds"**. In 2022 IEEE 29th International Conference on High Performance Computing, Data and Analytics Workshop (HiPCW), pp. 53-58. IEEE, 2022.

Professional Experience

Graduate Research Assistant, *August 2022 - May 2023, January 2024 - Current*
Rochester Institute of Technology, *Rochester, NY, USA*
- Performed original research on GPU resource management, I/O optimization, and autonomous vehicles.
- Published six peer-reviewed works (two as first author), including three full papers, two posters, and one workshop paper; selected entries above.

Graduate Teaching Assistant, *August 2023 - December 2023*
Rochester Institute of Technology, *Rochester, NY, USA*
- Resolved students' questions on class topics and assignments in office hours or online.

Senior Software Engineer, Software Engineer, *July 2018 - April 2022*
BJIT Ltd - Offshore Software Development Company, Dhaka, Bangladesh
- Developed custom computer vision models.
- Worked on Windows C++/CLR applications to deploy computer vision models on Intel edge devices using the OpenVINO toolkit.
- Trained newly recruited engineers on machine learning.
- Provided initial system design based on client requirements as a representative of a small team.

Skill Set

Programming Languages: C++, Python3, C, Bash, x86 Assembly; **GPU Runtime:** CUDA; **ML, CV Toolkit:** OpenCV, Scikit-learn, Keras, PyTorch, Hugging Face, OpenVINO; **Build systems**: Make, CMake, Bazel; **DBMS:** MySQL, SQLite, PostgreSQL; **H/W Development Platform:** Arduino, Raspberry Pi

Relevant Project Experience

GPU Context Shared ML Surrogate Workflow
- Proposed fusing training and inference context for continual fine-tuning to improve inference latency stability on a single edge GPU.
- Designed an algorithm to schedule training and inference processing in a unified context with best-effort deadline maintenance.
- Applied the developed approach to a deadline-constrained ptychographic restoration workflow, achieved a $13\% - 22\%$ deadline miss rate reduction vs multi-context baseline and reduced reconstruction MSE by up to 30% vs pre-trained baseline on two datasets.

Neural Network Parameter Offloading for Inference
- Implemented a performance model-based network parameter offloading to reduce GPU memory usage with a controllable performance penalty.
- Demonstrated $27\% - 52\%$ memory savings while incurring $1.1\times - 1.9\times$ slowdown through evaluation across networks for computer vision to LLM tasks.

Training Data Deduplication for Distributed Deep Learning
- Proposed a scheduling strategy for completely deduplicated In-memory Distributed Storage of Training Data, achieving $7\times$ data-loading I/O speedup compared to using the baseline PyTorch DDP setup.
- Designed a polynomial-time-complexity distributed storage update algorithm.
- Implemented the system as a dataloader to use with the PyTorch framework.

CUDA IPC Tensor Sharing for Co-Located DNN Workloads
- Implemented a tensor storage sharing mechanism using **CUDA IPC** in **PyTorch v1.13**.
- Implemented an `nn` module API to share the CUDA parameter set associated with the network through `/dev/shm` storage.
- Reduced the GPU memory usage by up to 40% in the scenario of co-located training and inference processes.

Tensor Engine Implementation
- Built a tensor engine with **C++** backend and **Python3** interface using **pybind11** as part of course work (GitHub Link).
- Implemented tensor storage and allocation for both CPU and GPU(**CUDA**).
- Implemented accelerated operations for CPU using multithreaded operations orchestrated by **OpenMP**.

Achievements and Activities

Artifact Review Committee Member *2024, 2025, 2026*
- Reviewed artifacts in ICPE'24, EuroSys'25,'26

ALCF Intro to AI-Driven Science on Supercomputers Digital Badge *December 2024*
- Workshop on AI in Argonne Leadership Computing Facility

ACM HPDC'23 Best Poster Award *June 2023*
- `https://twitter.com/ACM_HPDC/status/1671990446174642177/photo/1`

Part of BrickHack 9 Best Health Hack Winner Team *February 2023*
- RIT's premier collegiate hackathon